2020 IEEE International Conference on Computing, Power and Communication Technologies (GUCON)
Galgotias University, Greater Noida, UP, India. Oct 2-4, 2020

A Comparative Study of Data Mining, Digital Image Processing and Genetical Approach for Early Detection of Liver Cancer

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***Abstract*—Liver cancer is one of the leading causes of cancer deaths in Asia and Africa. The largest internal organ of our body is the liver. It performs many of the life sustaining functions and elementarily has effect on every physiological process occurring in our body. Liver is the second most organ which is metastatistically affected by cancer. Now a days, liver cancer becomes the main cause of death worldwide. At that time, the early prediction of liver cancer becomes necessary for increasing patient survival. In this paper, a comparison is made to know which method from genetic, data mining and digital image processing is best suited for early detection of liver cancer. After doing comparison, in this paper we decide to explore genetical approach. All the statistical methods work on microarray expression dataset and come under the category of genetical approach. Microarray data analysis evolved as an advanced research topic during recent decades. The key reason for analyzing microarray dataset is to identify and predict the gene that displays various values of expression under various testing conditions. The purpose of this research paper is to examine various forms of parametric statistical methods proposed to analyze data on microarray expression to classify the genes expressed differently and to investigate the mechanism of disease growth. Additionally, we have estimated the best situation for each system where they are doing better. In our research, we use the simulated datasets to thoroughly analyze and compare the various types of parametric methods.**

Keywords—data mining, digital image processing, parametric and non-parametric statistical methods, Liver cancer

I. INTRODUCTION

Now a day's liver cancer becomes the main cause of death worldwide and it is also referred to as hepatic cancer. Cancers that starts from liver is referred as primary liver cancers. Liver cancer is made up of malignant or benign growths found inside or over the liver. Hepatocellular carcinoma (or hepatoma or HCC) is found in majority in liver cancer and is affects more males as compared to females. Early detection of liver cancer becomes a main problem in liver cancer treatment. [1] Digital image processing is worked as a non-invasive technique for detecting tumors or cancers. Improvement in imaging techniques has made it possible to used it on high resolution CT image datasets, so that small tumors can be easily detected by physicians with large tumors. [2, 25] In genetical approach, microarray technology is utilized. The development of Microarray is being designed as a modern effective device in biotechnology. It has the capability to review of the expression profiles of more than 10000 genes at the same time. [3] Data mining technique used for data analyses such as Clustering, Classification and Association Rules and many more. Data mining also contribute the procedure and technology for doing data analysis and extract some useful information for decision making. [4]

In this paper, firstly three different approaches are studied which utilized for early detection of liver cancer such as Digital Image Processing, Data Mining Approach and Genetical approach include microarray expression data. After analyzing all the research papers, a comparative Table 1 is provided, summarizing the Digital Image Processing, genetical approach and Data mining approach.

Genetical approach in biotechnology offers a way to analyse the behaviour of thousands of gene on the basis of their expression profiles simultaneously by the use of hybridization technique. [5] Gene research expression profiles are important as they offer a way to recognize or predict markers of disease that are important to medical care. Many researchers would like to classify genes that are expressed differently in diseased condition. Such all methods have been divided into two groups, as parametric methods and non-parametric methods. The main purpose of this analysis is to check out the best methods that correctly classify the highest proportion of differential genes expressed with fewer percentages of equivalently expressed (EE) genes incorrectly included in differential genes expressed. [6] Researchers are able to identify only a small fraction of differentially expressed genes, by using statistical methods and it is the major problem during the selection of genes. The genes expressed differently are called important or discriminative genes. [7]

Numerous latest articles [5-10,11] are present which has provided the comparative study of different variety of statistical methods which works on microarray data. In this paper, we compare various parametric methods, and three different methods come under categories other than parametric and non-parametric methods such as Fold change, RAM and Rank product. Comparison of various non-parametric methods has been provided in [12]. In section 2, a review details are presented on each of these various statistical tests used for identifying genes which shows differential expressed patterns under various experimental conditions. A comparative research with consideration of various statistical methods outlined in Section 3. Statistical methods are employed on data sets which are explained in Section 4, and their results with discussion on different statistical methods are explained in Section 5. Finally, conclusion is presented.

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II. Various Statistical Tests for Predicting Differentially Expressed Genes

A. Fold Change

It is the most popular method because of its simplicity. In literature, two definitions for fold change are found. Standard definition [13] is the ratio of the means under two different experimental conditions. Fold change method is not performed well in selecting genes which shows different expression in microarray dataset, mainly under less abundant transcripts.

B. Parametric Tests

1) T-statistics

The most basic and fundamental statistical tests frequently used for identifying differentially expressed genes is two sample t-statistic. [14] One more test is known as Welch's test. [15] It is somehow similar to the t-statistic, but the only difference is that a known variance is used in denominator.

2) Bayes T-test or Bayesian method

For analyzing microarray data, a Bayesian probabilistic framework is developed by Baldi and Long. [16] This statistic is utilized to resolve the problem arises due to small variance under the less expressed values and utilizes the Bayesian statistical test in parametric form to estimate the parameters value such as standard deviation, mean etc. for T statistical test. It is more preferable and popular in analyzing the small sample size, because of its effectiveness under small sample size data.

3) B-statistics

A statistic is proposed by Lonnstedt and Speed [17] made on the basis of empirical Bayes approach by utilizing a Bayes log posterior odds. This statistic overcome T-statistic problem arises by log odds posterior statistic under normality assumption. Lonnstedt and Speed hierarchical model is implemented into a practical approach by Smyth [18] and named it B-Statistic. On the basis of general linear model, model is reset with random coordinates and contrast of interest.

4) ANNOVA-1 test

ANOVA-1(ANalysis Of Variance-1) [14] is a statistical test in which means of various groups or samples at different experimental conditions may or may not be equal for all genes due to this reason it is concluded that t-test work on greater than two groups. Performing this multiple two sample t-test result in a more probability of making type-1 error [19,20]. So due to this, ANOVA-1 test is more effective when we want to make comparison between means of two or more than two populations/groups. This method compares the means of two or more groups/population by comparing the estimated variance. The aim of doing this variance comparison is to know the mean difference between groups for creating statistical significance by analyzing variance. The basic of ANOVA-1 is that the two-independent estimation of population variance which computed from the sample dataset.

5) Pearson's Correlation Test (corr)

Correlation is a statistical measure of extent to two variables varied with respect to each other; it can be positively or negatively correlated. In correlation, assumption is made that data is from the bivariate population (involves two variables). Pearson correlation coefficient [21] is a way to represent the strength of linear association found between two variables.

C. Rank Product

The rank product [6] method is made to overcome the problem arises in fold change method, it is statistically both at the same time hard as well as simple [22]. After gaining the recognition, as a statistical test for identify genes which are differentially expressed from microarray data, Koziol [23] defines it in an extended form onto two sample data settings. This approximation is not performed well, if data has skewed.

D. Ranking analysis of Microarray (RAM)

RAM is a large scale two sample t-statistic [24], works on the basis of comparison made between ranked t-statistics set and ranked Z-values (one set is produced from ranked estimated null scores). These ranked Z-values are estimated using "random splitting" approach instead of permutation approach. RAM performs well under undesirable conditions like small sample size, large fudge factor and data containing numerous distribution of noises than SAM. [24]

III. Simulation Results and Discussion

After analysing all statistical methods used to analyze microarray data, with the exception of non-parametric methods, are presented in Table 2 which summarizes the statistical methods used with their best results, benefits and limitations. For performing experiments, we have generated simulated datasets are of 10 sample sizes having 1,000 genes under both the experimental conditions such as diseased or normal, assumes the same variance (such as std1=std2=2) but with different means $(\text{mean1} = \text{mean2} + 2)$. Datasets which follow Normal distribution utilized for both the conditions. We selected normal distribution patterns not only because they will generate the homogeneity/symmetry in data distribution for each condition, but also because in this research paper comparison includes the parametric statistical tests only. Our aim is to find differentially expressed genes accurately [26,27], for this purpose we used Matlab and R software's. Simulated datasets have been prepared for performing simulation. Then used filtering procedure to remove those genes who have low variance, for this purpose we used 'genevarfilter' Matlab function. Then we have utilized the parametric statistical tests, because these tests are well suited for normally distributed datasets. Corresponding p-value has been calculated for each and every gene by applying each one of statistical test. After that, Volcanoplot are utilized for predicting differentially expressed genes according to their ranks. An extensive simulation study has been done for evaluating the performance of each and every earlier method. We generated 1000 genes of different sample sizes (10, 20 and 40 samples) data. Fig.1 shows the simulation results with number of DEG's and Fig. 2, Fig. 3 and Fig. 4 shows the Volcanoplots having equal variance and different means.

IV. Conclusion

This paper has provided an comprehensive analysis of various parametric statistical studies, which are used to predict genes with distinct expression patterns under two experimental conditions. These parametric methods are evaluated by performing experiments by assuming p-value cutoff of 0.05 for the three simulated datasets. Statistical tests

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performance has been affected by few conditions like size of sample, data distribution, variance of data, genes number, etc. So, for getting the reliable and significant results, we first require to get information about the real characteristics of the given dataset and then employ the correct appropriate statistical method on that particular dataset. The traditional statistical tests applied on microarray data to identify differentially expressed genes, or a clustering algorithm to identify the groups of genes that shows similar behaviour. This methodology is not successful for predicting the genes co-expressed differentially [30] which have significant biological roles. Statistical test which have better performance to identify genes which shows co-expressed differential

TABLE I. SUMMARY OF ALL THE APPROACHES USED FOR EARLY DETECTION OF LIVER CANCER

Sr. no.	Genetical approach	Data mining Approach	Digital image processing
1	Depends on genes.	Depends on Risk factors.	Depends on images.
2	Works on microarray expression data.	Works on Survey data.	Works on digital image processing.
3	Measure difference in expression levels	Measure the probability of Risk factor present in an individual.	Measure differences in pixel intensity
4	Result in terms of differentially expressed genes.	Result of individual in terms of Susceptibility towards Liver cancer in future.	Result in terms of features
5	Problems come during the selection of genes	Problems come during the identification of Risk factor	Problems come during the segmentation of an image.
6	Utilizes statistical methods	Utilizes Data Mining techniques	Utilizes digital images processing techniques

TABLE II. COMPARISON OF DIFFERENT PARAMETRIC STATISTICAL TESTS USED TO ANALYSE MICROARRAY DATA

Test	Sample size	Data distribution	Variance	Advantages	Limitations
Fold change	Small	Both	Don't control	Popular because of its simplicity and easy to understand	This method not take variability into consideration
T-statistics	Large	Normally	Same or equal	Popular because of its simplicity and easy to understand	More error prone (high FDR) because of variance problem.
Welch's t	Large	Normally	Unequal variance	More robust than t-test; This test is simpler to use than other	This test should not be used on lognormal data. Under skewed authenticity decreases.
Bayes t-test or Bayesian	Small	Both	Assume unequal variance	Popular because of its effectiveness in evaluating samples size dataset	We cannot rely on any standard cutoff value for selection of DEG.
B-statistics	Large	Non-normal	Require equal variance	Popular because it analyze small sample size effectively	Difficult to estimate hyperparameters
ANOVA1 test	Large	Normal	Same variance	Useful where comparison is made in mean between two or more groups.	Inappropriate if the different columns represent different variables instead of different groups
Correlation Test	Poor in both	Poorer for both types	Do not control	Commonly used in linear regression	Not applicable for predicting differentially expressed gene/miRNA
Rank product	Small	Poor in skewed data distribution	Constant	Statistically simple and rigorous at the same time	Does not perform will if data is skewed
RAM	All size	Gamma and normally distribution	NA	Insensitive to the treatment effect that often presents in real data and having a better estimate of FDR It works well for small sample sizes.	Less number of permutated samples are needed for small sample size, which result to a biased ranking test and even gives the test not applicable.

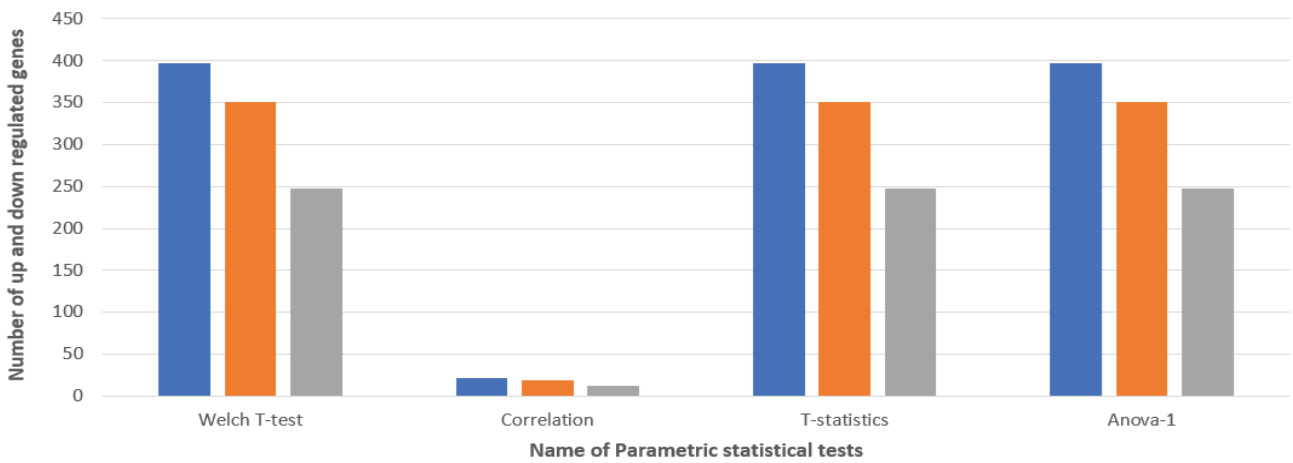

Fig. 1. Differentially expressed genes under different statistical test: In this Figure, Blue color represent the results comes in sample size 10, Orange color represent the results comes in sample size 20 and Grey color represent the results comes in sample size 40.

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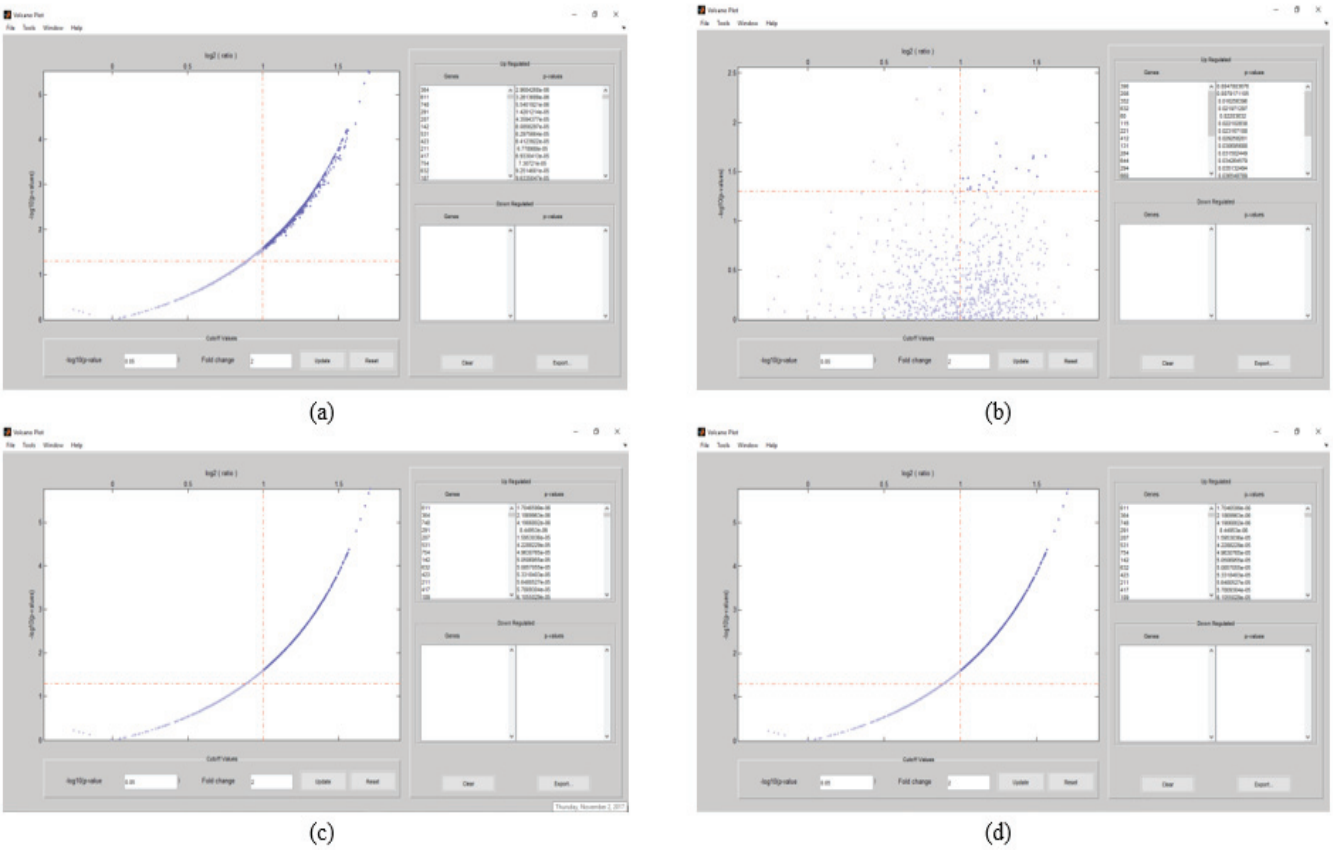

Fig. 2. Volcanoplot of predicted differential up and down regulated genes from the simulated datasets of sample size 10 by different types of statistical tests -(a) Welch T-test, (b) Correlation, (c) T-test, and (d) Annova1.

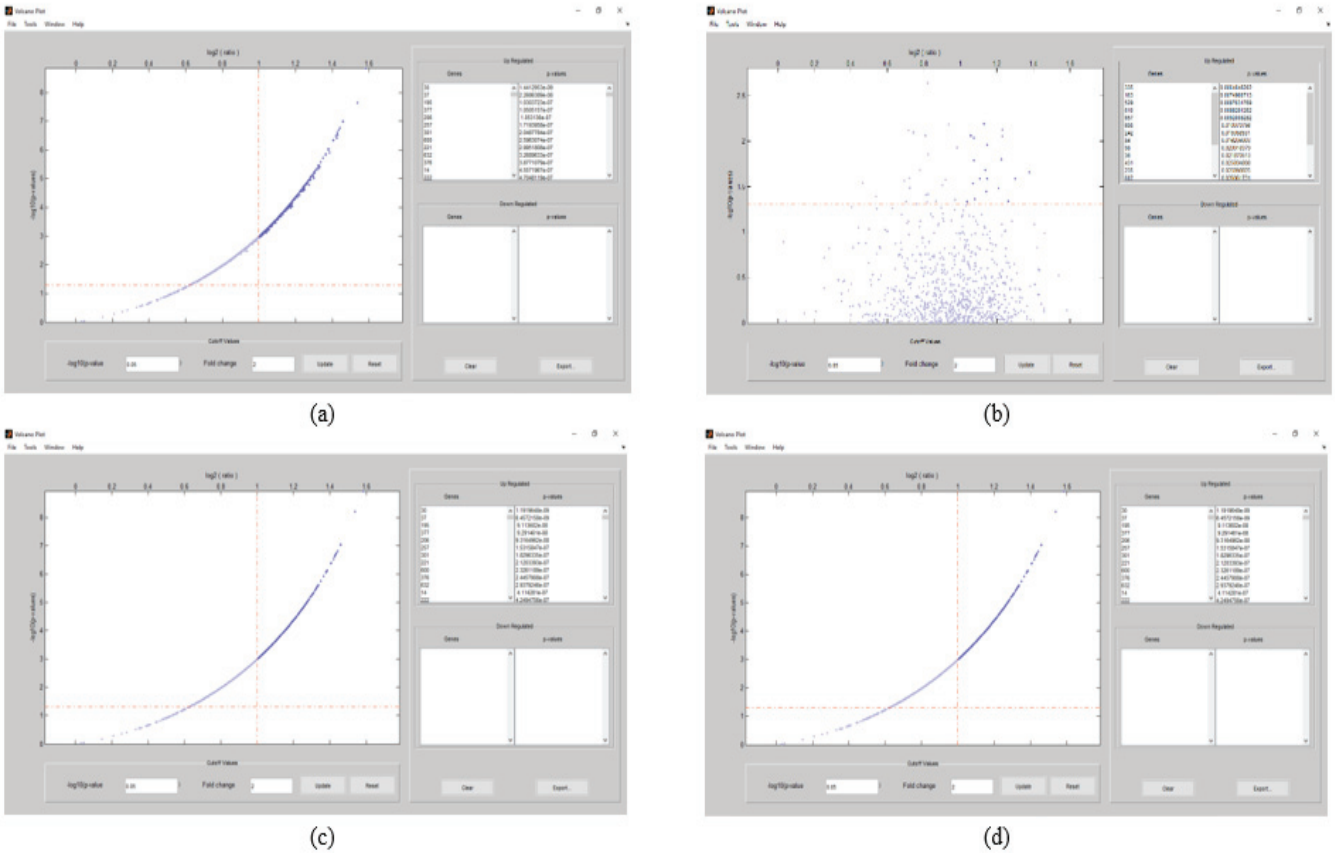

Fig. 3. Volcanoplot of predicted differential up and down regulated genes from the simulated datasets of sample size 20 by different types of statistical tests -(a) Welch T-test, (b) Correlation, (c) T-test, and (d) Annova1.

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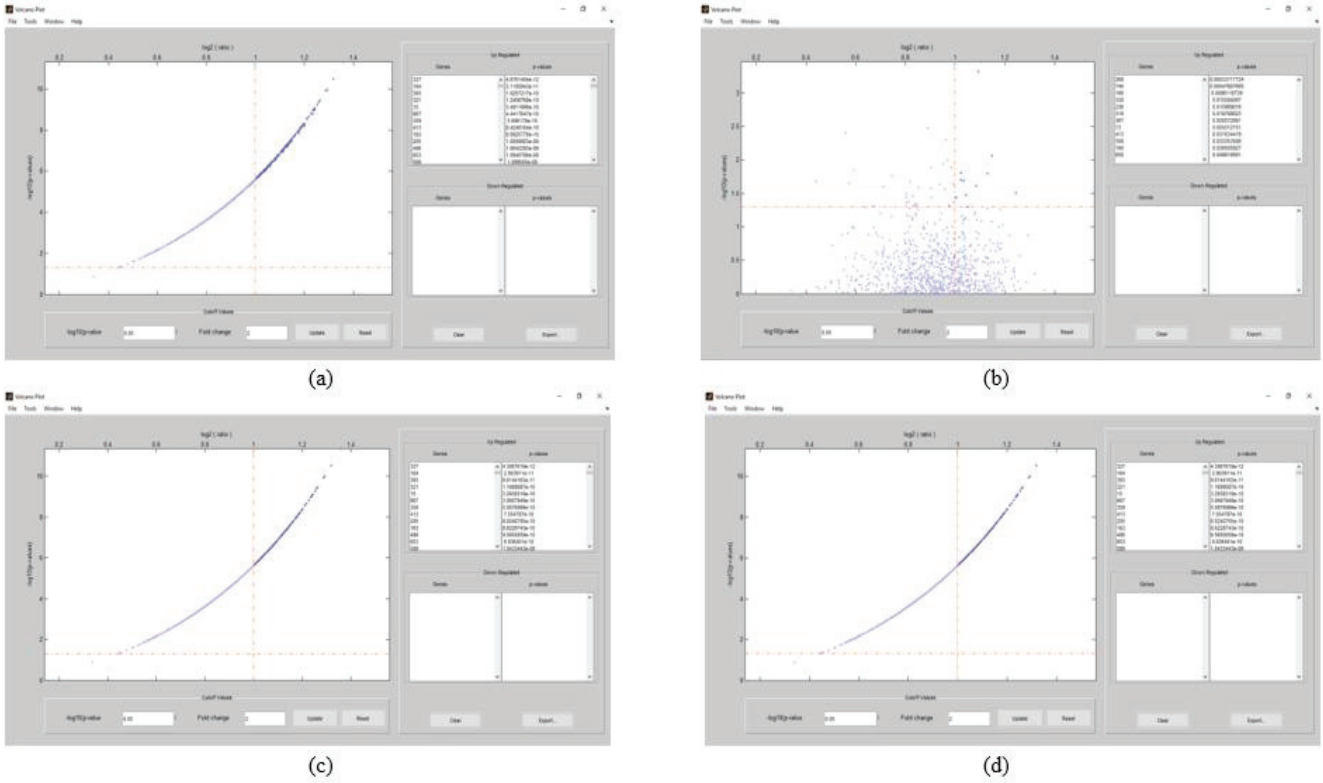

Fig. 4. Volcanoplot of predicted differential up and down regulated genes from the simulated datasets of sample size 40 by different types of statistical tests -(a) Welch T-test, (b) Correlation, (c) T-test, and (d) Annova1.

patterns are not many, for this reason further work is needed in this field.

ACKNOWLEDGEMENT

I am thankful to Indian Council of Medical Research (ICMR), New Delhi, India for providing me funds for doing this research.

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